Gregory Xie

Cambridge, MA | gregory.w.xie@gmail.com | gregoryxie.com |

Full-stack roboticist focusing on manipulator design, with experience spanning from mechanism design to modeling and control. Expert at optimizing high-performance robots by systems integration and tightly coupling electromechanical design to software and controls.

Experience

Roboticist | Google DeepMind | Cambridge, MA *May 2026 - Present*

Roboticist | Robotics and AI Institute | Cambridge, MA *Jun 2023 - May 2026*

Technical lead for the design of high-performance robotic hands and wrists for dynamic, bimanual manipulators, owning decisions from concepts through detailed drawings and integration.

- Designed underactuated and fully articulated tendon-driven hands (6-12 DoF) with integrated tactile sensing, enabling robust grasping and fine manipulation
- Designed a robotic forearm housing a compact 2-DoF parallel wrist and finger actuators, balancing strength, weight and packaging constraints. Optimized structure for weight using finite element analysis
- Designed rotary and linear quasi-direct drive finger and wrist actuators (20-40 mm diameter), enabling compliant yet forceful manipulation behaviors
- Developed models to study the effects of friction, mass, and latency on the dynamic performance of robot hands, informing actuator, transmission, and sensor design. Developed simulation tools to evaluate hand kinematics
- Performed workspace analysis and trajectory optimization for a bimanual manipulator, determining kinematic parameters, motor sizing, gearing, bearing selection, and power system requirements
- Created detailed part drawings using GD&T, performed tolerance analyses to ensure accurate and repeatable assemblies

Graduate Research Assistant | MIT CSAIL | Cambridge, MA *Sep 2022 - May 2023*
Robotics Software Engineering Intern | Realtime Robotics | Boston, MA *Jun 2022 - Aug 2022*
Mechatronics Intern | Nimble Robotics | San Francisco, CA *Jun 2021 - Aug 2021*
Mechanical Engineering Intern | Formlabs | Somerville, MA *Jun 2020 - Aug 2020*

Patents and Publications

G. Xie. "Compact Wrist Actuation Mechanism for a Robotic Arm." U.S. Provisional Patent Application No. 63/907,403, Oct. 2025.
G. Xie, N. Rojas. "Wrist Mechanism for a Robot Arm." U.S. Patent Application No. 19/043,948, Feb. 2025.

A. S. Morgan, **G. Xie**, et al. "AthenaZero: A Low-Inertia, Bimanual Robot for Dynamic Manipulation." *Science Robotics*, 2026.
L. Chin, **G. Xie**, J. Lipton, D. Rus. "Large-Expansion Bi-Layer Auxetics Create Compliant Cellular Motion." *IEEE ICRA*, 2025.
G. Xie, L. Chin, B. Kim, R. Holladay, D. Rus. "Strong Compliant Grasps Using a Cable-Driven Soft Gripper." *IEEE IROS*, 2024.
G. Xie, R. Holladay, L. Chin, D. Rus. "In-Hand Manipulation with a Simple Belted Parallel-Jaw Gripper." *IEEE RA-L*, 2024.
L. Chin, M. Burns, **G. Xie**, D. Rus. "Flipper-Style Locomotion through Strong Expanding Modular Robots." *IEEE RA-L*, 2023.

Education

Massachusetts Institute of Technology

M.Eng. in Electrical Engineering and Computer Science (GPA: 5.0/5.0) *2022 - 2023*
B.S. in Electrical Engineering and Computer Science and B.S. in Mechanical Engineering (GPA: 5.0/5.0) *2018 - 2022*

Skills

Robotics	Robot kinematics, modeling, computational design optimization, MuJoCo, Drake
Programming	Python, C++, MATLAB
Mechanical and Electrical	CAD (SolidWorks, Onshape), FEA (Ansys), GD&T, DFM/DFA, KiCad, LTspice
Fabrication	CNC and manual machining, 3D printing, laser and waterjet cutting, soldering